An Interpretable Multi-Modal Spam Detection Framework Using Explainable AI and Machine Learning

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***Abstract*—Scam messages reach users through several channels, yet the machine-learning filters that intercept them usually return a verdict without a reason. We present an interpretable multi-modal framework that detects fraudulent SMS and e-mail messages and explains every decision. Each channel has a specialist pipeline that maps URLs, phone numbers and money amounts to readable placeholder tokens, combines TF–IDF features with fifteen stylometric statistics, and fuses Logistic Regression, Random Forest and XGBoost by F1-weighted soft voting; LIME explains the full ensemble and exact TreeSHAP the boosted component. On held-out splits of the UCI SMS Spam Collection and the Enron-Spam corpus, the ensemble reaches 98.55% accuracy (ROC-AUC 0.9941) on SMS and 98.25% (ROC-AUC 0.9983) on e-mail. Paired McNemar and DeLong tests show that it significantly outperforms both tree learners on e-mail ($p \leq 0.003$) and ties Logistic Regression, and a word-deletion test confirms that LIME's explanations are faithful. Global attributions show that stylometric features dominate SMS decisions and expose a corpus shortcut: the token `enron` occurs in 42% of legitimate test e-mails and in no spam, and appending it to spam e-mails flips 35.5% of the ensemble's detections (91.7% for XGBoost alone).**

***Index Terms*—Spam detection, smishing, phishing, explainable artificial intelligence, SHAP, LIME, ensemble learning, shortcut learning, stylometry.**

I. Introduction

Fraudulent messaging is the most frequently reported form of cybercrime: in 2025 the FBI's Internet Crime Complaint Center logged more than one million complaints and US$20.9 billion in losses, with phishing and spoofing again the largest category (191,561 complaints) [1]. Large language models (LLMs) now let attackers write persuasive, personalised phishing at negligible cost [2], [3], and scams have followed banking and one-time-password traffic onto SMS [4].

Machine-learning filters intercept most of this traffic, but they are usually black boxes: a message is blocked and nobody can say why. High accuracy can hide shortcuts, spurious cues that happen to separate the classes in one dataset [5], and although explainable AI (XAI) now offers a broad toolkit [6], explanations in spam filtering are mostly shown as illustrations rather than tested for faithfulness [7] or used to audit the model. Most detectors also serve a single channel, although SMS and e-mail differ sharply (a median of 12 versus 155 words and 12% versus 48% spam in the corpora used here), and comparisons on near-saturated benchmarks rarely report uncertainty.

We address these gaps with an interpretable multi-modal framework, where *modality* denotes the messaging channel (SMS or e-mail). Our contributions are:

1) a CPU-only pipeline with one specialist per channel: readable placeholder tokens, TF–IDF plus fifteen named stylometric features, and Logistic Regression (LR), Random Forest (RF) and XGBoost (XGB) fused by F1-weighted soft voting;
2) a statistically grounded evaluation with bootstrap confidence intervals, McNemar and DeLong tests, and an end-to-end evaluation of channel routing and cross-channel transfer;
3) a dual explanation layer (LIME for the full ensemble, exact TreeSHAP for XGBoost) whose faithfulness is tested by word deletion; and
4) an explanation-driven audit that uncovers and quantifies a corpus shortcut, and a sparse-versus-dense pitfall that silently invalidates SHAP explanations of XGBoost text models.

II. Related Work

SMS spam and smishing. Reproducible SMS research rests on the UCI SMS Spam Collection [8]. Recent detectors are

mostly deep or transformer models, such as a smishing detector designed to reduce false positives [4] and a transformer detector with an explainability analysis [9]; Hur *et al.* found LLMs robust and effective for SMS spam [10]. Şahin *et al.* benchmarked classical learners across text representations on the same corpus and found the representation more decisive than classifier complexity [11]. Work on other languages such as Swahili is growing [12], and Salman *et al.* released a new dataset and analysed spammers' evasive techniques [13].

E-mail spam and phishing. The Enron-Spam corpus [14] remains a standard e-mail benchmark. Fine-tuned transformers and LLMs dominate recent work [15]–[18]; Meléndez *et al.* found transformers ahead of traditional models for phishing e-mail, by a small margin [19], and meta-learner ensembles remain competitive [20]. Hadi *et al.* target both channels with a deep–shallow hybrid [21]. LLMs can author effective phishing at scale [2], and Opara *et al.* showed that such messages can bypass major webmail filters but are detectable with stylometric features [22].

Explainability and shortcuts. LIME fits a local surrogate around one prediction [23]; SHAP assigns additive Shapley attributions [24], computed exactly for trees by TreeSHAP [25]. Explanations have accompanied smishing detectors [26], [27], guided feature selection for spam [28], and been compared on boosting models, where SHAP was more stable globally than LIME [29]. They are not trustworthy by default: faithfulness must be measured [7], explainers often disagree [30], and many challenges remain open [6]. Shortcut learning, the exploitation of spurious dataset cues, is well documented in NLP [5], [31]; spam corpora are vulnerable because legitimate mail often comes from a single organisation.

Table I contrasts representative recent studies. Most address one channel and use explanations mainly as illustrations. We combine two channel specialists with a dual explanation layer, test its faithfulness, use it to audit the model, and compare learners with paired significance tests.

III. Proposed Framework

Fig. 1 shows the architecture. A message is routed to the specialist of its channel; each specialist has its own normalisation, feature space and ensemble, and returns a label, a spam probability and explanations.

A. Normalisation and Features

Text is lower-cased, HTML tags are stripped, and high-signal entities are *mapped* rather than deleted: URLs become `httpaddr`, e-mail addresses `emailaddr`, currency amounts `moneysymb`, phone numbers `phonenumbr` and other numbers `numbr`. These are among the strongest scam signals, and as readable tokens they can appear in an explanation; no stemming is applied, so every explained token is a recognisable word. The text is vectorised with sublinear TF–IDF over uni- and bi-grams (3,000 terms with $\mathrm{df} \geq 2$ for SMS, 5,000 with $\mathrm{df} \geq 3$ for e-mail). Fifteen stylometric statistics computed on the *raw* message (Table II) are scaled by their training maximum and appended, giving 3,015 and 5,015 dimensions; their upper-case names keep them distinct from vocabulary in every explanation.

B. Base Learners and Weighted Soft Voting

Three learners with different inductive biases provide diversity: LR (liblinear, $C = 1$) and RF [32] (250 trees, minimum leaf size 2) with balanced class weights, and XGBoost [33] (300 trees, depth 6, learning rate 0.1, subsampling 0.9/0.8) with $\mathrm{scale_pos_weight} = N_{\mathrm{ham}}/N_{\mathrm{spam}}$. Re-weighting was preferred to synthetic oversampling so that every training example remains a real, explainable message. The ensemble averages the learners' probabilities,

$$p(\mathrm{spam} \mid x) = \sum_{m=1}^{3} w_m \, p_m(\mathrm{spam} \mid x), \tag{1}$$

with w_m proportional to each learner's mean F1 in stratified 5-fold cross-validation on the training split, and predicts spam when $p \geq 0.5$. The weights are 0.3329/0.3331/0.3340 (LR/RF/XGB) for SMS and 0.3356/0.3311/0.3333 for e-mail, so any benefit comes from diversity rather than from a dominant member.

C. Channel Router

A transparent rule selects the specialist: a message is treated as e-mail if it exceeds 280 characters, contains a line break or contains a cue word (*subject*, *dear*, *regards*, *sincerely*, *unsubscribe*, *attached*), and as SMS otherwise. Users can override the choice in the application.

D. Explanation Layer

LIME [23] removes words from the raw message, re-scores 1,000 perturbations through the *entire* pipeline and fits a local linear surrogate, so it explains the complete ensemble through its ten most influential words. For XGBoost, SHAP values decompose the log-odds output additively, $f(x) = \phi_0 + \sum_j \phi_j(x)$ [24], and are computed exactly by TreeSHAP [25]; global importance is the mean $|\phi_j|$ over 800 held-out messages. One implementation detail is critical: XGBoost treats entries absent from a sparse matrix as *missing*, whereas a densified copy turns them into zeros that follow different branches. Densifying the input before explaining therefore explains a different function: it changes the XGBoost verdict for 729 of 1,032 SMS and 1,182 of 2,400 e-mail test messages. All attributions here use XGBoost's native TreeSHAP on the sparse matrix the model actually scores.

IV. Experimental Setup

The UCI SMS Spam Collection [8] is used in full, and a class-stratified sample of 12,000 messages (subject and body) is drawn from the Enron-Spam corpus [14], after removing empty and duplicate messages (Table III). Each corpus is split 80/20 with stratification (seed 42); all fitting, including the cross-validated ensemble weights, uses the training split only, and the test split is used once. We report accuracy, precision, recall, F1, MCC, ROC-AUC, PR-AUC and the Brier score, with 95% bootstrap confidence intervals (2,000 resamples).

TABLE I
REPRESENTATIVE RECENT STUDIES COMPARED WITH THIS WORK

Study	Year	Channel	Approach	Explanation
Salman *et al.* [13]	2024	SMS	ML/DL models under evasive techniques; new dataset	–
Roumeliotis *et al.* [16]	2024	E-mail	Fine-tuned LLMs, NLP models and CNNs	–
Meléndez *et al.* [19]	2024	E-mail (phishing)	Traditional ML vs. transformers	–
Uddin *et al.* [9]	2025	SMS	Transformer language model	Post-hoc explainability analysis
Kshirsagar *et al.* [20]	2025	E-mail	Meta-learner ensembles	Interpretability-oriented
Uddin *et al.* [17]	2026	E-mail (phishing)	Fine-tuned transformer	Post-hoc explanations
Koide *et al.* [18]	2026	E-mail (phishing)	LLM-based detection	LLM-generated rationale
Jasim *et al.* [27]	2026	SMS	Fuzzy logic + BiLSTM	Fuzzy rules
Prince *et al.* [26]	2026	SMS (smishing)	Hybrid transformer feature fusion	Post-hoc XAI
Hadi *et al.* [21]	2026	SMS + e-mail (evaluated on SMS)	Shallow lexical + deep contextual features	–
This work	2026	SMS + e-mail (routed specialists)	TF–IDF + stylometry; LR, RF, XGB fused by F1-weighted soft voting	LIME (full ensemble) + exact TreeSHAP; deletion faithfulness test; shortcut audit

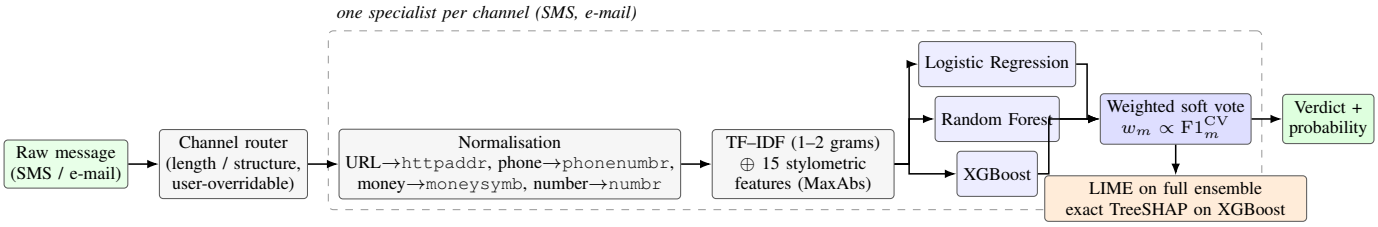


Fig. 1. Architecture of the proposed framework. Each channel has its own normalisation, feature space and ensemble; the explanation layer returns a local explanation with every verdict and global summaries of each model.

TABLE II
ENGINEERED STYLOMETRIC FEATURES (COMPUTED ON THE RAW MESSAGE)

Feature	Description
`CHAR_COUNT`, `WORD_COUNT`	Length in characters / words
`AVG_WORD_LEN`	Mean word length
`DIGIT_RATIO`	Share of characters that are digits
`UPPERCASE_RATIO`	Share of letters in upper case
`ALLCAPS_WORDS`	Words written fully in capitals
`PUNCT_RATIO`	Share of punctuation characters
`SPECIAL_CHAR_RATIO`	Share of non-alphanumeric characters
`NUM_EXCLAMATIONS`	Count of "!"
`NUM_QUESTIONS`	Count of "?"
`NUM_CURRENCY`	Currency symbols and money amounts
`NUM_URLS`, `NUM_EMAILS`	Detected links / e-mail addresses
`NUM_PHONES`	Detected phone numbers
`SPAM_KEYWORDS`	Hits in a 46-word, human-auditable scam lexicon

TABLE III
CORPORA AFTER CLEANING AND DE-DUPLICATION

Channel	Messages	Spam	Median words	Train	Test
SMS [8]	5,158	642 (12.4%)	12	4,126	1,032
E-mail [14]	12,000	5,792 (48.3%)	155	9,600	2,400

The ensemble is compared with each base learner on the same messages using the exact McNemar test [34] and DeLong's test for correlated ROC curves [35], Holm-corrected within each channel.

We further evaluate (i) the router on the pooled test sets (3,432 messages) and each specialist on the other channel; (ii) LIME faithfulness, by deleting the top-k LIME words ($k \in \{1, 3, 5\}$) from 60 SMS and 25 e-mail correctly detected spam messages and comparing the drop in spam probability with that from deleting k random words; and (iii) the shortcut surfaced by the global attributions, by deleting the token from legitimate test e-mails and inserting it into spam. The framework is implemented in Python with scikit-learn, XGBoost, SHAP and LIME and served through a four-tab Streamlit application; a 13-check smoke test covers the whole pipeline, and re-running training reproduces all metrics exactly. Timings use a 10-core ARM64 CPU without a GPU.

V. RESULTS

A. Classification Performance and Significance

Table IV and Fig. 2 report all four systems. On SMS the ensemble reaches 98.55% accuracy with the highest precision (0.9669), only 4 false positives among 904 legitimate messages and the best PR-AUC and Brier score; on e-mail it reaches 98.25% with the highest recall (0.9940; 7 of 1,158 spam messages missed) and ROC-AUC (0.9983). The SMS specialist favours precision, which suits a channel where a blocked message may be a one-time password; the e-mail specialist favours recall. LR is a strong baseline: it matches

TABLE IV
HELD-OUT TEST PERFORMANCE WITH 95% BOOTSTRAP CONFIDENCE INTERVALS (SMS: $n = 1{,}032$; E-MAIL: $n = 2{,}400$). LOWER BRIER IS BETTER.

Channel	Model	Accuracy	Precision	Recall	F1	MCC	ROC-AUC	PR-AUC	Brier
SMS	LR	0.9855 [.978, .992]	0.9520 [.911, .984]	0.9297 [.882, .970]	0.9407 [.910, .967]	0.9325 [.898, .963]	0.9941 [.989, .998]	0.9768 [.959, .992]	0.0148 [.011, .019]
	RF	0.9845 [.977, .991]	0.9516 [.905, .985]	0.9219 [.869, .965]	0.9365 [.902, .964]	0.9278 [.890, .959]	0.9929 [.985, .998]	0.9770 [.958, .992]	0.0184 [.015, .023]
	XGB	0.9845 [.977, .991]	0.9667 [.928, .992]	0.9062 [.849, .954]	0.9355 [.902, .963]	0.9273 [.890, .959]	0.9941 [.989, .998]	0.9739 [.954, .989]	0.0135 [.008, .021]
	Ensemble	**0.9855** [.978, .992]	**0.9669** [.929, .992]	**0.9141** [.860, .960]	**0.9398** [.907, .967]	**0.9320** [.895, .962]	**0.9941** [.988, .999]	**0.9790** [.961, .993]	**0.0130** [.009, .018]
E-mail	LR	0.9833 [.978, .988]	0.9753 [.966, .984]	0.9905 [.984, .996]	0.9829 [.978, .988]	0.9668 [.957, .977]	0.9979 [.997, .999]	0.9976 [.996, .999]	0.0210 [.018, .024]
	RF	0.9746 [.968, .980]	0.9629 [.951, .973]	0.9853 [.978, .992]	0.9740 [.967, .980]	0.9494 [.937, .961]	0.9973 [.996, .998]	0.9970 [.996, .998]	0.0380 [.035, .041]
	XGB	0.9750 [.969, .981]	0.9645 [.954, .975]	0.9845 [.977, .991]	0.9744 [.968, .981]	0.9502 [.938, .963]	0.9969 [.996, .998]	0.9965 [.995, .998]	0.0189 [.015, .023]
	Ensemble	**0.9825** [.977, .988]	**0.9705** [.961, .980]	**0.9940** [.989, .998]	**0.9821** [.977, .987]	**0.9652** [.955, .975]	**0.9983** [.997, .999]	**0.9980** [.997, .999]	**0.0213** [.019, .024]

TABLE V
PAIRED TESTS: ENSEMBLE VS. BASE LEARNERS. b/c: MESSAGES ONLY THE ENSEMBLE / ONLY THE BASE LEARNER CLASSIFIES CORRECTLY. p-VALUES HOLM-CORRECTED.

		McNemar		DeLong (ROC-AUC)		
Ch.	**Comparison**	b/c	p	ΔAUC	z	p
SMS	Ens. vs. LR	4/4	1.000	+0.0000	0.02	1.000
	Ens. vs. RF	2/1	1.000	+0.0012	1.15	0.754
	Ens. vs. XGB	1/0	1.000	-0.0000	-0.01	1.000
E-mail	Ens. vs. LR	11/13	0.839	+0.0004	1.26	0.208
	Ens. vs. RF	24/5	0.002	+0.0009	3.14	0.003
	Ens. vs. XGB	22/4	0.002	+0.0014	3.66	$<$0.001

TABLE VI
ROUTING AND CROSS-CHANNEL TRANSFER OF THE ENSEMBLES (POOLED TEST SET: 3,432 MESSAGES)

Setting	Acc.	F1	MCC	AUC	FPR
Oracle routing (pooled)	0.9834	0.9780	0.9648	0.9978	0.0182
Automatic routing (pooled)	0.9216	0.8921	0.8317	0.9687	0.0443
SMS model $\rightarrow$ e-mail test	0.5208	0.5287	0.0442	0.5290	0.5129
E-mail model $\rightarrow$ SMS test	0.2209	0.2299	0.0594	0.6707	0.8805

TABLE VII
DELETION TEST OF LIME FAITHFULNESS: MEAN DROP IN ENSEMBLE SPAM PROBABILITY AFTER REMOVING THE TOP-k LIME WORDS / k RANDOM WORDS

Channel	$k = 1$	$k = 3$	$k = 5$	Flips (k=5)
SMS ($n = 60$)	0.106 / 0.012	0.291 / 0.043	0.455 / 0.081	52%
E-mail ($n = 25$)	0.043 / 0.002	0.133 / 0.005	0.198 / 0.010	24%

the ensemble's SMS accuracy and slightly exceeds its SMS F1 and e-mail accuracy.

The paired tests (Table V) resolve these small differences. On SMS none is significant. On e-mail the ensemble beats both tree models (McNemar: 24 vs. 5 discordant messages against RF and 22 vs. 4 against XGB, $p = 0.002$; DeLong $p \leq 0.003$) but is statistically indistinguishable from LR (11 vs. 13, $p = 0.84$).

B. Routing and Cross-Channel Transfer

The specialists do not transfer (Table VI): the SMS model is at chance on e-mail (ROC-AUC 0.53), and the e-mail model flags 88% of legitimate SMS as spam. The router sends 96.4% of SMS and 85.1% of e-mails to the right specialist, but misrouted e-mails (short, without cue words) are classified correctly only 42.3% of the time, so pooled accuracy falls from 98.34% with oracle routing to 92.16% with automatic routing. The heuristic router, not the classifiers, is the weakest component, hence the manual override.

C. Local and Global Explanations

Fig. 3 explains an illustrative smishing SMS written by the authors and absent from both corpora: "URGENT! You have WON a £2,000 cash prize GUARANTEED. Call 09061701461 now or visit www.claim-your-prize.com to claim!" All learners vote spam (ensemble 0.996). LIME ranks the premium-rate number highest, followed by the money amount and scam vocabulary; TreeSHAP attributes the decision mainly to stylometry (digit ratio, word length, lexicon hits, capitals) and the `httpaddr` and `phonenumbr` placeholders. LIME speaks in the user's words, SHAP in the model's features.

Globally (Fig. 4), eight of the ten most important SMS features are stylometric, and the fifteen engineered features (0.5% of the dimensions) carry 74.5% of the attribution mass: a short scam is betrayed by its *shape* as much as by its words. On e-mail a single word, `enron`, dominates with about three times the attribution of the next feature.

D. Faithfulness of the Local Explanations

On SMS, deleting the top LIME word lowers the ensemble's spam probability 8.6 times more than deleting a random word (Table VII); deleting the top five lowers it by 0.455 and flips 52% of the verdicts. E-mail drops are smaller in absolute terms, because long messages contain redundant evidence, but about 20 times larger than random. LIME therefore points at words on which the decision genuinely depends.

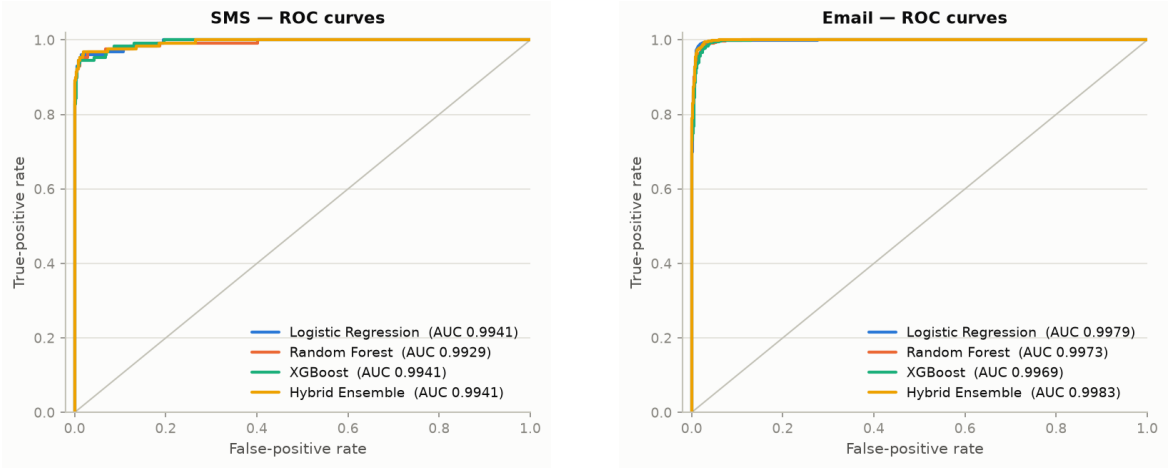
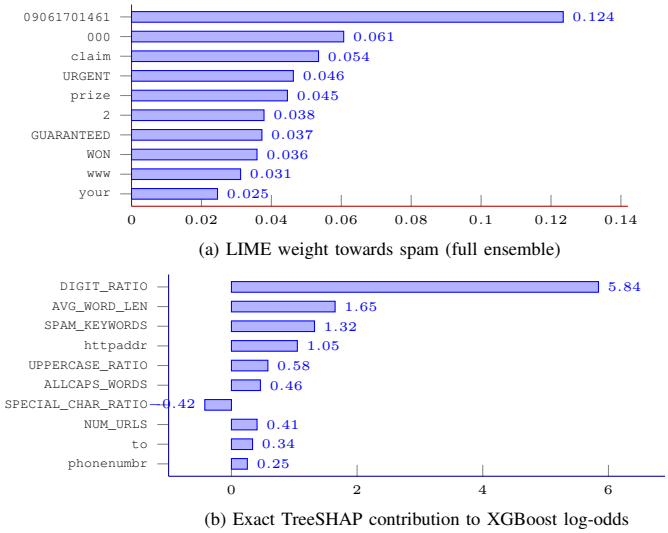


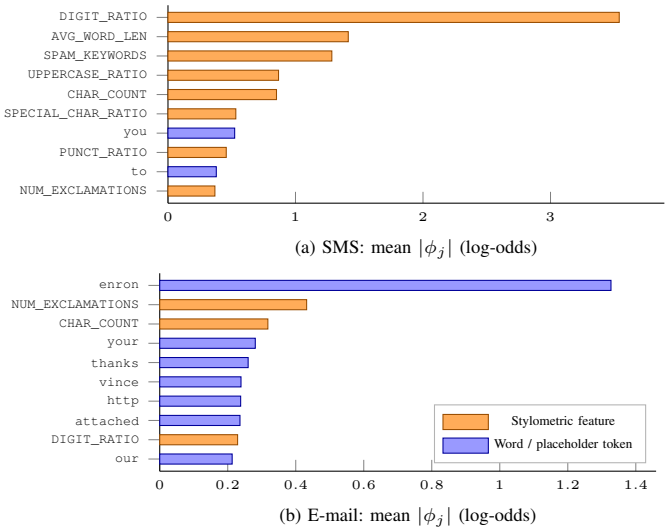
Fig. 2. ROC curves of the four systems on the held-out SMS (left) and e-mail (right) test sets. All systems exceed 0.99 AUC; differences are resolved by the paired tests in Table V.



(a) LIME weight towards spam (full ensemble)

(b) Exact TreeSHAP contribution to XGBoost log-odds

Fig. 3. Local explanations of the illustrative smishing SMS (ensemble spam probability 0.996): (a) LIME weights over the full ensemble; (b) the ten largest exact TreeSHAP contributions of the XGBoost component (base value 0.50 log-odds). Upper-case names are stylometric features.



(a) SMS: mean $|\phi_j|$ (log-odds)

(b) E-mail: mean $|\phi_j|$ (log-odds)

Fig. 4. Global importance (exact TreeSHAP, XGBoost component, 800 held-out messages per channel): the ten most important features for (a) SMS and (b) e-mail.

E. Explanations as an Audit Tool: A Corpus Shortcut

The token `enron` occurs in 44.4% of legitimate training and 42.1% of legitimate test e-mails but in *no* spam: every legitimate message is internal Enron correspondence, so the company name is a one-sided proxy for ham. Its mean TreeSHAP contribution is -3.95 log-odds when present versus $+0.66$ when absent; other identity cues such as the first name *vince* also rank highly, and none of the ensemble's 35 false positives contains the token. Table VIII quantifies the risk. Deleting the token from 523 legitimate e-mails flips only 11 verdicts, but *appending* the single word to spam flips 409 of 1,151 detections (35.5%). XGBoost alone is evaded on 91.7% of messages, RF on 25.3% and LR on 4.5%, so the heterogeneous ensemble is far more robust than its boosted member, yet not immune. The shortcut is invisible in Tables IV and V; only the explanations revealed it.

TABLE VIII
SHORTCUT PROBE ON THE E-MAIL TEST SET: INSERTING "ENRON" INTO DETECTED SPAM AND DELETING IT FROM LEGITIMATE MAIL

	Insertion into spam		Deletion from ham	
Model	**Flips to ham**	$\overline{\Delta p}$	**Flips**	$\overline{\Delta p}$
LR	52/1147 (4.5%)	-0.066	2/523	+0.032
RF	289/1141 (25.3%)	-0.301	10/523	+0.053
XGB	1045/1140 (91.7%)	-0.763	19/523	+0.052
Ensemble	409/1151 (35.5%)	-0.376	11/523	+0.046

TABLE IX
COMPUTATIONAL COST ON A 10-CORE ARM64 CPU (NO GPU)

Channel	Train (s)	Verdict (ms)	Batch (msg/s)	SHAP (ms)	LIME (ms)
SMS	14.3	14.7	14,897	0.86	76
E-mail	76.8	14.8	1,967	0.75	210

F. Computational Cost

Training both specialists, including cross-validation, takes about 1.5 minutes on a CPU (Table IX). A verdict takes about 15 ms, a TreeSHAP attribution under 1 ms and a LIME explanation 0.08–0.21 s, so every prediction can be explained interactively.

VI. DISCUSSION AND LIMITATIONS

All learners exceed 97% accuracy, and a regularised linear model on well-designed features is as accurate as the ensemble, consistent with the finding that representation matters more than classifier complexity for SMS spam [11]. The ensemble's value lies in better ranking metrics, a precision-first SMS profile, significant gains over both tree models on e-mail and, above all, robustness to the shortcut attack. The `enron` shortcut is a property of the corpus rather than the pipeline, yet a black-box deployment would have shipped it silently. Global explanations are therefore an auditing instrument, but they must themselves be validated [7]: the sparse/dense pitfall yields plausible SHAP plots of a function the deployed model never computes. We recommend computing tree attributions on the exact input the model scores, validating explanations by deletion, screening global attributions for corpus identifiers, and evaluating routing as well as classification.

The corpora are English and date from 2000–2012, so modern and code-mixed smishing and LLM-generated phishing [22] are not covered, and the shortcut means e-mail accuracy overstates performance on other organisations' mail. Results come from one stratified split, the router is heuristic, SHAP covers only the boosted component, and neither transformer or LLM detectors nor adversarial obfuscation [13] were evaluated.

VII. CONCLUSION

We presented an interpretable multi-modal framework that detects scam SMS and e-mail with channel-specific ensembles of LR, RF and XGBoost and explains every verdict with LIME and exact TreeSHAP. It reaches 98.55% (SMS) and 98.25% (e-mail) accuracy, significantly outperforms both tree learners on e-mail and trains in minutes on a CPU. Its explanations are faithful, show that stylometry rivals vocabulary on SMS, and exposed a corpus shortcut that a one-word insertion can exploit. Future work will test cross-corpus generalisation and remove identity shortcuts, extend the framework to multilingual and code-mixed data including Indian languages, learn the router, and benchmark transformer and LLM detectors under the same protocol.